F_U_Bi_i Seven-Component Force Decomposition: Phonon, Inflation, BCS, VDS, DVP, BSH, and QCalcGeom Sectors

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PAPER_1088: $F_{U,Bi,i}$ Seven-Component Force Decomposition

Abstract

The unified buoyancy force $F_{U,Bi,i}$ is decomposed into seven orthogonal sectors that account for all fundamental buoyancy-mediated interactions in the UQFF framework:

$$F_{U,Bi,i} = F_{\text{phonon}} + F_{\text{inflation}} + F_{\text{BCS}} + F_{\text{VDS}} + F_{\text{DVP}} + F_{\text{BSH}} + F_{\text{QCalcGeom}}$$

Each sector contributes a fractional share to the total buoyancy budget measured against the base gravitational acceleration $g_{extbase} = \mu_s \nabla(M_s/r)$.

§1 Base Gravity and Component Definitions

$$g_{\text{base}} = \frac{GM}{r^2}, \quad G = 6.674 \times 10^{-11}\ \text{m}^3/(\text{kg} \cdot \text{s}^2)$$

The seven force components are:

§1.1 Phonon Sector

$$F_{\text{phonon}} = \Phi_{1.25\,\text{THz}} \cdot g_{\text{base}} \cdot M$$

Vacuum phonon resonance at 1.25 THz, coupling buoyancy to the SCm lattice oscillation spectrum.

§1.2 Inflation Sector

$$F_{\text{inflation}} = \beta_i \cdot U_g \cdot \frac{M}{d} \cdot [\text{UA}]$$

Buoyancy force driving inflationary expansion, with $\beta_i = 0.603$ and $[\text{UA}] = 10^{-4}$.

§1.3 BCS (Buoyancy-Condensate-Superfluid) Sector

$$F_{\text{BCS}} = \Delta^2_{\text{BCS}} \cdot g_{\text{base}} \cdot M$$

Cooper-pair condensate contribution from the BCS gap parameter Δ_{BCS}.

§1.4 VDS (Vacuum Density Series) Sector

$$F_{\text{VDS}} = \rho_{\text{vac}} \cdot V_{\text{eff}} \cdot g_{\text{base}}$$

Vacuum density series contribution encoding $k_\eta = 10^{-113}$.

§1.5 DVP (Dipole Vortex Primes) Sector

$$F_{\text{DVP}} = \sum_{p \in \mathcal{P}} \frac{\mu_p}{r_p^3} \cdot g_{\text{base}} \cdot M$$

Vortex-line contributions indexed by prime-number dipole configurations.

§1.6 BSH (Buoyancy Shell Harmonics) Sector

$$F_{\text{BSH}} = \sum_{\ell=0}^{26} Y_\ell^0(\theta) \cdot g_{\text{base}} \cdot M$$

Spherical-harmonic shell decomposition across 26 layers.

§1.7 QCalcGeom (Quantum Calculator Geometry) Sector

$$F_{\text{QCalcGeom}} = \alpha_{\text{QCG}} \cdot g_{\text{base}} \cdot M$$

Geometric correction from the quantum calculator manifold structure.

§2 Fractional Contribution Table

Sector	Force Component	Fraction of $F_{U,Bi,i}$
Phonon	F_{phonon}	Dominant at resonance
Inflation	$F_{\text{inflation}}$	$\beta_i / \sum$
BCS	F_{BCS}	$\sim 10^{-3}$
VDS	F_{VDS}	$\sim 10^{-5}$
DVP	F_{DVP}	$\sim 10^{-4}$
BSH	F_{BSH}	$\sim 10^{-2}$
QCalcGeom	$F_{\text{QCalcGeom}}$	$\sim 10^{-3}$

§3 Budget Closure

$$\sum_{k=1}^{7} f_k = 1, \quad f_k \equiv \frac{F_k}{F_{U,Bi,i}}$$

The seven fractions are computed dynamically per system and must satisfy exact budget closure to machine precision.

References

- PAPER_1089: Inflation Buoyancy Sector Lagrangian
- PAPER_1090: Dark Energy Buoyancy Sector Lagrangian
- PAPER_892: FUBiUnifiedFieldBuoyancyForceCalc
- PAPER_893: GravBuoyancyForceCalculator